

Qatar University

Department of Mathematics and Statistics

STAT 482 – Bayesian Statistics — Spring 2026

Student Paper Presentation – Guidance Sheet Paper 2

Faster Estimation of Bayesian Models in Ecology Using Hamiltonian Monte Carlo

Monnahan, C.C., Thorson, J.T., & Branch, T.A.

Methods in Ecology and Evolution, 8, 339–348 (2017).

DOI: [10.1111/2041-210X.12681](https://doi.org/10.1111/2041-210X.12681)

[Open Access](#)

How to use this guide. This is a *methodological* paper – its main contribution is explaining and benchmarking the HMC algorithm. Unlike the other papers, Step 2 is not about a specific data model but about the *algorithm* itself. You are expected to understand **why** Stan uses HMC, and to demonstrate a working RStan model in your replication.

#	Component	Time	What the audience expects to see
1	Problem & Motivation	~2 min	Why is MCMC efficiency a real problem? What fails in JAGS?
2	HMC – the “Model”	~3 min	Hamiltonian dynamics, NUTS; intuition with equations
3	Benchmarking Method	~2 min	Metric used; models tested; experimental design
4	Key Findings	~3 min	Efficiency ratios; when Stan wins; NCP result
5	Your Replication	~3 min	A Stan model you ran; ESS and diagnostics you checked
6	Lessons Learned	~2 min	What you now understand about RStan from the inside

Step 1 – Problem and Motivation

~2 minutes

- Q1.** The paper says JAGS can be “prohibitively slow” for some models. Can you give a concrete example of a model type where this is a real problem in ecological research? Why does model complexity affect sampling speed?
- Q2.** What is **random walk behaviour** in an MCMC sampler? Use your own words and explain why it leads to high autocorrelation and slow mixing.
- Q3.** Look at Figure 1 in the paper (citation trends for Stan vs. the BUGS family). What does the trend suggest about the direction the field is moving? Why do you think adoption of Stan was slow in ecology at the time of publication?
- Q4.** In STAT 482 we use RStan for all our computations. After reading the introduction of this paper, can you explain to a classmate *why* the course uses Stan rather than JAGS?

Step 2 – HMC: The Algorithm

~3 minutes

This step replaces the “Bayesian model” step for this paper. You must explain HMC clearly, with a diagram or analogy, and write at least one equation.

- Q1.** The paper introduces the concept of a ball rolling on a frictionless surface. In this analogy:
- What does the *position* of the ball correspond to in the MCMC context?
 - What does the *potential energy* correspond to?
 - What does the *kinetic energy* correspond to?
 - Why does the total energy (the Hamiltonian H) remaining constant help produce good MCMC proposals?
- Q2.** Write the Hamiltonian as a function of position θ and momentum $\mathbf{p}$:

$$H(\theta, \mathbf{p}) = U(\theta) + K(\mathbf{p})$$

What are U and K in terms of the posterior distribution $\pi(\theta | \mathbf{y})$?

- Q3.** What is the **leapfrog integrator**? Why must a numerical approximation be used instead of the exact Hamiltonian path? What two tuning parameters control the leapfrog trajectory?
- Q4.** What is the **No-U-Turn Sampler (NUTS)** and how does it solve the trajectory length problem of static HMC? What does “U-turn” mean geometrically?
- Q5.** HMC requires the *gradient* of the log-posterior. What technique does Stan use to compute this gradient automatically? Why is this important for practical usability?

Key Concept: Why HMC avoids random walk behaviour

Think about what happens when the MCMC chain is in a region of low posterior probability far from the mode. A Gibbs sampler takes small steps and will take many iterations to return to the high-probability region. How does the HMC “ball” behave in the same situation? What property of the dynamics prevents it from getting stuck?

Step 3 – Benchmarking Method

~2 minutes

- Q1.** What metric do the authors use to compare Stan and JAGS? Define **effective sample size (ESS)** precisely: how does it differ from the raw number of MCMC iterations, and why is the ratio ESS / time a better measure of efficiency than raw computation speed?
- Q2.** The paper tests models of varying size and complexity. Name two ecological models used in the benchmark and describe their structure briefly. Why is a hierarchical model a harder case for MCMC than a simple regression?
- Q3.** What is **non-centered parameterization (NCP)**? If a random effect is written as $\alpha_j \sim \mathcal{N}(\mu, \tau^2)$, how would you re-write this in non-centered form? Why can the centering choice dramatically affect Stan’s efficiency?

Step 4 – Key Findings

~3 minutes

- Q1.** For simple, small models: what do the results say about the relative efficiency of Stan versus JAGS? Is this surprising?
- Q2.** For larger, more complex models: by approximately how much does Stan outperform JAGS (in terms of ESS/min)? What feature of the model drives the biggest gains?

- Q3.** For hierarchical models: the paper finds that NCP is important for Stan but less so for JAGS. Can you explain why HMC is *more* sensitive to parameterization than Gibbs sampling?
- Q4.** The paper also discusses **divergent transitions** – a diagnostic unique to HMC. What is a divergent transition geometrically? Why does Stan warn about them while JAGS gives no equivalent warning? What should you do when you see divergent transitions in your Stan output?
- Q5.** What are the stated **limitations** of Stan identified in the paper? Think about situations in STAT 482 where these limitations would matter.

Step 5 – Your Replication

~3 minutes

R / RStan Exploration Hints

Fit a Bayesian regression or population model in RStan and use it to illustrate the diagnostics and efficiency concepts from the paper.

- **Suggested model:** a simple logistic growth model for population size, or a Bayesian hierarchical model with random effects. Both appear in the paper and are directly relevant to STAT 482.
- **What to examine:** after fitting, check `get_sampler_params(fit)` for divergent transitions (`divergent_`). Count them per chain. Try increasing `adapt_delta` from 0.8 to 0.95 – does the number decrease?
- **Measure ESS:** use `summary(fit)$summary[, 'n_eff']` and compare to the total number of post-warmup draws. What fraction of your draws are effectively independent?
- **NCP experiment:** fit a hierarchical model with centered parameterization (e.g. $\alpha_j \sim \mathcal{N}(\mu, \tau^2)$ directly) and then re-fit with NCP ($\alpha_j = \mu + \tau z_j$, $z_j \sim \mathcal{N}(0, 1)$). Does ESS change? Do divergences disappear?
- **Useful functions:** `check_divergences(fit)`, `check_treedepth(fit)`, `pairs(fit, pars=...)`, `ggs_traceplot()` from `ggmcmc`.

- Q1.** When you look at the pairs plot (`pairs(fit, pars=...)`) for a hierarchical model, what pattern indicates the “funnel” geometry problem? Can you see it in your output?
- Q2.** After applying NCP, does your trace plot look better? How do you define “better” in this context?

Step 6 – Lessons Learned

~2 minutes

- Q1.** Before this paper, you may have used RStan as a “black box.” After reading it, what do you now understand about what Stan is actually doing inside the `stan()` call? Name three things you now know that you did not before.
- Q2.** The paper was published in 2017. Stan has evolved considerably since then. Do you think the conclusions still hold? What would you test to update the benchmark today?
- Q3.** When you run `stan()` in STAT 482 and Stan reports “X divergent transitions after warmup,” what will you do now that you have read this paper?
- Q4.** The paper focuses on ecology. Can you think of a Bayesian model from another domain – economics, medicine, climate science – where the same HMC efficiency gains would be critical?

- Do not describe HMC as “just a faster version of MCMC.” The key insight is the use of *gradient information* to make directed proposals – explain this clearly.
- Do not skip the NCP discussion. It is one of the most practically important results of the paper for STAT 482 students.
- If Stan produces divergent transitions in your replication, do not ignore them. Discuss what they indicate and how you tried to address them.
- Do not confuse $\hat{R}$ (a convergence diagnostic) with ESS (an efficiency measure). They answer different questions.

Suggested Slide Structure (9 slides)

- S1.** Title, paper reference, your name
- S2.** Motivation: why JAGS is slow; random walk illustrated
- S3.** Hamiltonian dynamics: ball-on-surface analogy with diagram
- S4.** HMC algorithm: leapfrog integrator, NUTS stopping criterion
- S5.** Benchmark setup: models tested, ESS/min metric defined
- S6.** Results: efficiency table or figure – Stan vs. JAGS
- S7.** Non-centered parameterization: before and after comparison
- S8.** Your replication: Stan output, divergence check, ESS ratio
- S9.** Lessons learned: what you now understand about Stan